Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/01/2005
PI Name:	Merfeld, Dan Ph.D.		
Project Title:	Influence of Sensory Integration on the Neural Processing of Gravito-Inertial Cues		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Boston	**State:**	MA
Zip Code:	02114	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	12/01/2003	**End Date:**	03/31/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is 3/31/2007 per PI (11/06). NOTE: Funding reinstated to 11/30/2006, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Lionel, Zupan Ph.D. (MEEI)		
Grant/Contract No.:	NNJ04HB01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Our sense of spatial orientation results from a complex set of neural processes of sensory integration; these processes utilize information from many different physiological systems. Evidence suggests that extended exposure to micro-gravity yields adaptive changes in these neural processes. For the most part, these adaptive changes appear functionally relevant for spaceflight but yield significant transient problems when astronauts return to gravitational environments (Earth, Mars, Moon, etc.). In-flight countermeasures that reinforce the normal sensory interactions encountered in a gravitational environment (e.g., rotations about any axis not aligned with gravity are accompanied by changes in the relative orientation of gravity) will almost certainly help astronauts function adequately when they transition to a gravitational environment. This will be important to pilot astronauts landing spacecraft as well as to crewmembers trying to rapidly exit a spacecraft. Furthermore, artificial gravity has been proposed as a countermeasure for numerous problems associated with spaceflight (including cardiovascular deconditioning, bone decalcification, etc.). One undesired side effect of artificial gravity will be additional alterations in these processes of sensory integration. For all of these reasons, it is essential that we understand the predominant sensory interactions that are altered by exposure to micro-gravity. Specifically, we proposed to investigate how rotational signals from the semicircular canals influence how humans process the ambiguous gravito-inertial force measured by the otolith 0rgans. We are now performing a series of ground experiments that seek a better understanding of sensory interactions between the semicircular canals and graviceptors. We propose to perform these investigations using several different measurement techniques including measures of reflexive eye responses as well as psychophysical measures of human perception.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In our studies, we are investigating how normal subjects process ambiguous sensory cues from the otolith organs, which measure gravity and linear acceleration. These basic science studies will lead to improved diagnostic testing for human patients suffering from balance and vestibular problems.
Task Progress:	Our progress is documented in full in our 8-page progress report. In brief, we have published 4 papers in peer-reviewed journals with a 5th paper in press. These papers provide detailed descriptions of our scientific findings. On the technical front, we have also made substantial progress. Specifically, we have installed a new motion device, a variable radius tilt device, that we plan to use for all of our testing in the next year. We have also worked with our institution to design and build a new 1200 square foot facility that houses this new motion device as well as two additional new motion devices that should be installed and operational in the next 6 months.
Bibliography Type:	Description: (Last Updated: 04/05/2011)
Articles in Peer-reviewed Journals	Merfeld DM, Park S, Gianna-Poulin C, Black FO, Wood S. "Vestibular perception and action employ qualitatively different mechanisms. I. Frequency response of VOR and perceptual responses during Translation and Tilt." J Neurophysiol. 2005 Jul;94(1):186-98. PMID: 15728767 , Jul-2005
Articles in Peer-reviewed Journals	Merfeld DM, Park S, Gianna-Poulin C, Black FO, Wood S. "Vestibular perception and action employ qualitatively different mechanisms. II. VOR and perceptual responses during combined Tilt&Translation." J Neurophysiol. 2005 Jul;94(1):199-205. PMID: 15730979 , Jul-2005
Articles in Peer-reviewed Journals	Zupan LH, Merfeld DM. "An internal model of head kinematics predicts the influence of head orientation on reflexive eye movements." J Neural Eng. 2005 Sep;2(3):S180-97. PMID: 16135883 , Sep-2005
Articles in Peer-reviewed Journals	Poon C, Merfeld D. "Internal models: the state of the art. Editorial/introduction to special issue." J Neural Eng. 2005 Sep; 2(3): S144-8. http://www.iop.org/EJ/article/1741-2552/2/3/E01/jne5_3_e01.pdf , Sep-2005